Mobile network transmission quality

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Mobile communications systems and the environments in which they are used pose difficult challenges when ensuring the system provides acceptable transmission quality. Many of the key parameters affecting the speech transmission quality are determined by the design of the mobile terminal. In addition, the use of echo control devices and the error performance of the fixed links (within the mobile network as well as the interconnected networks), can have a major effect. The design of the speech codec also has a fundamental effect on the speech quality offered by the system.

This paper considers the various issues affecting mobile system speech quality. The design of the terminal will be discussed with particular reference to operational experience gained with phase 1 GSM terminals and recent standards activities related to handset and hands-free systems. Aspects of system design contributing to the performance are discussed, including the correct use of echo control devices, end-to-end planning standards and interconnect agreements, the susceptibility of the mobile network signalling systems to fixed link errors, and the effect on perceived quality. In addition, the paper will examine the opportunities for enhancing the current GSM system with new speech codecs, including the development of an enhanced full rate (EFR) speech codec. Particular attention is paid to the G.729 EFR submission, developed by BT Laboratories (BTL) and Cellnet, and the opportunity that this solution offers for a greatly improved radio error performance.

1. Introduction

Explosive growth of the mobile communications market in recent years has led to a variety of terrestrial mobile technologies. These technologies include the analogue cellular systems, such as TACS, AMPS and NMT and the digital cellular systems, such as DAMPS and CDMA in the USA, and GSM in Europe, as well as the DCS1800 and PCS1900 GSM derivatives. In addition, there are a number of digital cordless systems available such as DECT and CT2. Satellite systems, such as Inmarsat Standard M and Iridium, have also been evolving to offer mobile communications services with wide area coverage. The Appendix contains a full explanation of the acronyms used in this paper.

In particular, the GSM digital cellular system, together with its derivatives, has become a global success, offering mobility and ISDN services to customers in Europe, Asia-Pacific and the United States. Certain aspects of the GSM system can be enhanced to build on this success, taking GSM from a second generation digital cellular system to a communications medium, able to offer 'wire-line' quality to customers, whether they require mobility or not.

The mobile systems listed above incorporate a variety of functions which, if not designed and planned properly, can seriously degrade the speech quality of the end-to-end connection. Mobile terminals are frequently used in conditions of high ambient noise, and, because of the long propagation delay introduced by the speech and channel codecs, have to exhibit good acoustic-echo loss performance. These two aspects of the terminal's operation place difficult design criteria on equipment designers who have also to meet demands for small, lightweight terminals.

Speech and channel codecs, which are used to compress the amount of transmitted speech data and to protect that speech data in the potentially difficult radio environment, make a fundamental contribution to the system's quality. For GSM, these coding techniques have evolved from the initial GSM full-rate (FR) codec to today's half-rate (HR) and enhanced full-rate (EFR) codecs. Indeed, technology already exists which can offer a digital cellular customer 'wire-line' or PSTN speech quality across a broad range of radio conditions.

There are a number of elements in the fixed part of the mobile network that, if not planned properly, can degrade the speech quality. The echo control devices, used to protect the mobile customer from echo, can interact with other devices in the connection. Fixed link errors can affect the mobile speech frames and, more importantly, the signalling channels. The usual transmission parameters of delay, level and quantisation distortion, all require careful planning to ensure the customer receives acceptable quality calls across the wide variety of call types.

Within the United Kingdom, a new set of guidelines has been produced by the Public Network Operators Interest

Group (PNO-IG), a sub-group of the Department of Trade and Industry's (DTI) Network Interfaces Co-ordination Committee (NICC), to assist network operators in apportioning these parameters between networks to ensure end-to-end speech quality and assist in developing interconnect agreements. The UK Network Performance Design Standard (NPDS) [1] assigns the transmission parameters according to network type, ITU-T and ETSI specifications.

2. Terminal design

Many of the key speech quality parameters are determined solely within the GSM mobile. In addition, several of these parameters are critical to the interworking between the mobile network and any interconnected networks.

The send loudness rating (SLR), in the mobile-to-land direction, and the receive loudness rating (RLR) in the land-to-mobile direction, determine the audio signal levels for the customer's speech. The loudness ratings are calculated from the send and receive sensitivity masks or frequency responses. These are dictated by the acoustic transducers used as well as the anti-aliasing and reconstruction filters for the analogue-to-digital converters. One criticism levelled at GSM is that it is 'quiet'; this is difficult to understand as the SLR and RLR are in line with ITU-T long-term values [2]; however, it is important to remember how the SLR and RLR are measured. The terminal is mounted on a loudness rating guard-ring position (LRGP) artificial head with the ear cap sealed to the artificial ear. The artificial mouth position is adjusted such that the terminal's microphone is at the mouth reference point (MRP), i.e. 25 mm in front of the lip ring. The test is then carried out and the loudness ratings are calculated.

The problem lies with the design of the terminal equipment. Most mobile terminals are the shape and size of small chocolate bars and position the microphone transducer half way up the user's check. The air path between the microphone and the user's mouth introduces an additional acoustic loss of some 6 dB, raising the SLR from the nominal 8 dB to a figure of 14 dB, effectively reducing the send path audio level by 6 dB; this could be counteracted in the GSM transcoder (TRAU), in the fixed network, by varying the gain. However, the TRAU would introduce a fixed gain and the mobiles are all of different sizes and use different transducers. A fixed correction is therefore inappropriate.

Similar problems exist in the receive path. The mobile handset is mounted on the LRGP and the ear cap sealed to the artificial ear. If a seal cannot be achieved, the test house can use a 'rubber ring' to seal the ear; this means that although all the receive signal power from the ear piece is directed at the artificial ear, in the operational environment, however, there is an ear leak resulting in a modified acoustic response. Also, environmental noise can leak under the ear cap, masking the receive signal.

The audio levels and the shape of the handset also affect the echo loss of the mobile [3]. The echo loss or terminal coupling loss (TCL) protects the interconnected network from echo returning from the GSM network. The echo is perceptible due to the inherent delay of the GSM system. A weighted terminal coupling loss (TCL_w) of 46 dB is required to provide echo protection across the range of call scenarios from national and international mobile to PSTN calls and mobile-to-mobile calls. This figure is derived from the ITU-T G.131 [4] echo tolerance curves. These curves show that to provide echo protection for delays up to 300 ms an echo loss of 56 dB is required. It can be assumed that the far-end terminal has an overall loudness rating of 10 dB for planning purposes which means the GSM terminal has to achieve an echo loss of 46 dB. The echo path is dominated by the acoustic coupling between the earpiece and the microphone. As has already been stated, many mobile terminals place the microphone alongside the user's cheek. Ideally the transducers should be placed next to the ear and mouth. Unfortunately this dictates either a larger terminal, which consumers do not want, or a flip-phone design, which reduces the scope for manufacturers to differentiate their products.

It is important that the test method for TCL as well as the performance requirement are carefully considered. Echo problems were reported when GSM was first introduced, even on national calls. The GSM phase 1 TCL test allowed sinusoidal test stimuli to be passed through the speech codec, as some manufacturers use aspects of the speech codec in their acoustic echo cancellation devices; however, the GSM full-rate codec causes spectral spreading on sinusoidal signals which results in lower power when measured at their discrete frequencies. Speech signals are of wider bandwidth and transient, and are transmitted satisfactorily [5]; this meant that mobiles were passing the test but failing in an operational environment. The phase 2 test, developed by BT Laboratories (BTL), uses the ITU-T P.50 [6] artificial voice as the test stimulus to address this problem. The BTL proposal originally utilised a free space TCL_w measurement with the aim to move forward to a head and torso simulator (HATS) based system. In this measurement the terminal 'dangles' in free space within a quiet room during the measurement. The method was not acceptable to the terminal manufacturers and hence the standard reflects an LRGP-based test but with a revised test stimulus.

The terminal's volume control effects the TCL, as it increases the receive audio signal level. There is some debate as to whether the TCL should be decreased when the volume control is operated, since a higher ambient noise, usually the reason for increasing the volume, will mask the echo. Subjective testing carried out at BTL has shown that a TCL of 32 dB is still required when in ambient noise conditions; however, noise is not the only reason to turn up the volume control. A quiet far-end talker can also cause this action which changes the TCL without any noise masking to compensate. In addition the RLR is measured using a sealed ear while in an operational environment there is no seal and signal power is lost from under the ear cap. In addition the binaural human has to contend with ambient noise input to the other ear. GSM TCL has thus to be met at all positions of the volume control for handsets, but can be reduced for hands-free systems.

Acoustic echo cancellation can be used by the terminal designer as part of the TCL solution; however, to produce stable acoustic echo cancellation an adaptive algorithm is required because of the variable nature of the echo path. Such an algorithm introduces additional processing delay which potentially contributes to other end-to-end speech quality problems. A low-delay algorithm may not be as resilient but can be used as one part of the TCL solution.

The positioning and choice of the transducers has a large effect on the noise rejection capabilities of the terminal. There are two main noise rejection problems — noise leaking under the ear cap into the user's ear can mask the received speech, and noise coupled into the send path via the microphone can mask the mobile user's speech and adversely effect the operation of DTX and other network devices, such as speech interpolation systems and echo suppressers. The two parameters which control these paths are the listener sidetone (LSTR) and DELSM respectively. As with TCL, these problems are easier to control with good acoustic design practice and noise-cancelling microphones can also improve noise rejection performance [7].

Algorithmic solutions for noise reduction can be used but these tend to affect the basic speech quality and require additional delay budget to allow sufficient processing time, thus affecting the end-to-end delay and hence quality. Measurements carried out at BTL [8] on a number of mobile terminals have shown that small terminals can have very good noise rejection capabilities. The results ranged from −4 dB to +4 dB for single figure DELSM. This means that one mobile was 4 dB more sensitive to noise than speech while for another, the situation was reversed.

The interaction of acoustic noise with other system components can have a damaging effect on their ability to meet the needs of the customers. A particular example of this is the current half-rate codec which is known to be disturbed by background noise, resulting in a 'swirl' effect. An acoustic noise rejection mask has been demonstrated [9] to improve the performance greatly, but has been rejected by the ETSI standards group as it places stringent design restrictions on mobile terminals. BTL examined three noise rejection masks for the half-rate codec and these are shown in Fig 1.

Subjective assessment revealed that the low mask did not remove enough noise to prevent the codec's 'swirl' effect for all background noise levels while the high mask, although removing all the noise, unmasked the metallic nature of the half-rate codec's speech quality. The medium mask removed sufficient noise to virtually eradicate the 'swirl' effect while keeping the metallic nature of the codec largely masked. An alternative algorithmic approach has been proposed but this represents a significant additional development and is likely to affect the speech quality. Performance in background noise can have a serious effect on the public's perception of the service offered. The original CT2 telepoint non-common air interface (CAI) systems were heavily criticised for their noise performance as customers could not communicate while in the street, exactly the environment at which this service was being aimed. It is likely that poor noise performance was one factor in the demise of CT2 telepoint in the United Kingdom.

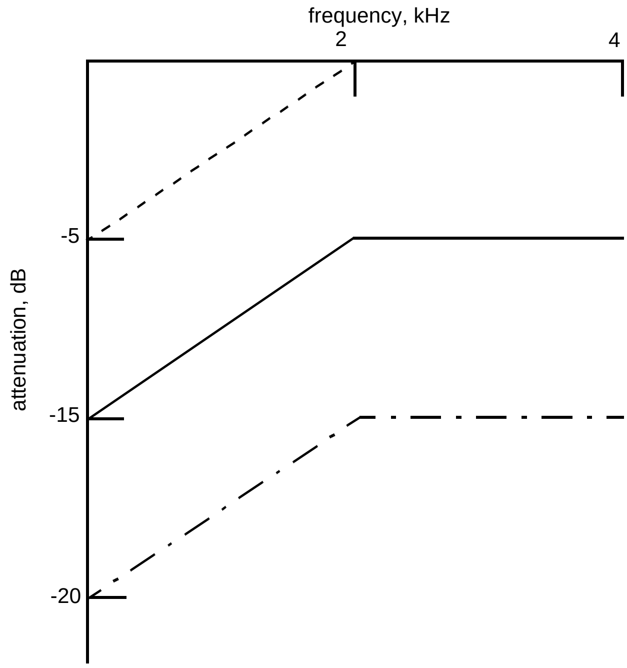


Fig 1 Proposed noise rejection masks for the GSM half-rate codec.

Hands-free terminals introduce further requirements in order to avoid compromising speech quality. The vehicle mounted hands-free environment is particularly hostile. The acoustic space is surrounded by a reflective material with high ambient noise levels. To overcome this the hands-free system has to use some algorithmic noise reduction and echo cancellation systems. These introduce additional delay, potentially degrading conversation quality. The recent adoption, within ETSI, of an additional 39 ms for hands-free processing has increased the GSM one-way delay budget by 40% and is likely to cause problems with end-to-end quality. In addition a TCL of 40 dB was adopted for hands-free. Again, the argument for reducing the TCL is that the noise will mask any echo; however, the GSM hands-free car phone is installed in an environment where several people can use it simultaneously. While normally positioned for the benefit of the driver, a rear seat passenger may wish to use it, or the passengers may all wish to contribute to a call, requiring the volume to be increased so that they can all hear the system. In addition, GSM systems are now emerging which are integrated into personal computers; such systems, combined with hands-free telephony could be used for conference calls in quiet environments, requiring additional volume with no noise masking to hide the echo. GSM also has the opportunity to address the needs of the wireless multimedia market and it is important that terminal manufacturers and network operators work together to ensure that the speech quality of the end-to-end connection is not compromised.

Current tests used for handset systems do not represent the operational environment of the terminal as they use traditional sinusoidal test stimuli and the speech codec is bypassed even though it is a potential source of degradation in the GSM system. In addition the handsets are mounted on an LRGP artificial head in sealed ear condition. To make testing more relevant to the operational environment it would be better to develop further test methods that use a head and torso simulator (HATS) which more accurately represents a human user. It is also suggested that tests should be more system oriented, using an artificial speech stimulus, such as the ITU-T P.50 [6] signal. Inclusion of the codec would also be more appropriate.

ETSI have now started a work package on examining how the GSM handset tests can be adapted to use more appropriate test stimuli and mounting systems. It is important that these new tests are incorporated in the handset type approval tests if they are to result in improved terminal performance from commercially available hand portables.

Currently, the minimum performance that a GSM cellular network operator can expect from terminals connected to the network is that they have passed the type approval tests. The tests set out in TBR 9 (GSM Phase 1) [10] and TBR 20 (GSM Phase 2) [11] are, quite rightly, a subset of the full GSM 11.10 [12] test suite. It is not necessary to test all the various quality and interworking parameters, but it is necessary to test the critical ones accurately, particularly those that effect the interconnect agreements with other network operators and other third-party obligations. The tolerances for the type approval tests in TBR 9 and 20 have been revisited by the standards committees, mainly due to the legal implications of failing these tests, and it is perhaps time for the GSM community to come together to re-examine the type-approval system, ensuring that critical tests are carried out both in an operationally relevant way and to ensure that the process leads to enhanced GSM products and services without proving a barrier to the terminal manufacturers.

3. Speech coding

One of the most fundamental parts of the GSM system is the speech codec. To attain the required spectral efficiency, the speech codec is used to compress the speech data to minimise the channel bit rate. To achieve this, the speech codec effectively models the vocal tract of the user as a filter and sends the filter coefficients to the decoder in the TRAU together with some residual excitation energy. This basic method enables the speech codec to reduce the speech data rate to 13 kbit/s for the full-rate (TCH-FS) and EFR codecs and 5.6 kbit/s for the half-rate (TCH-HS) codec. Further information on the GSM radio interface can be obtained in Hodges [13].

The reduction in data rate adds distortion to the speech signal, the extreme effect of which can be heard in some of the military speech codecs, where the emphasis has been placed on intelligibility rather than speaker recognition. As a result the speech sounds ‘robotic’. The traditional measure for this distortion is the quantisation distortion unit or QDU. One QDU is equivalent to the amount of distortion introduced by a single transition from analogue speech to 64 kbit/s G.711 [14] PCM and back to analogue speech. GSM recommendation 06.10 [15] states that the GSM speech codec introduces between 7 and 8 QDUs under error-free (EP0) radio conditions. The QDU is an accurate measure for tandem PCM and ADPCM systems, allowing planners to determine the amount of distortion in a call-routeing path and ensure that it does not breach ITU-T recommended limits. ITU-T G.113 [16] states that an international connection should not exceed 14 QDUs — this is broken down into the 5-4-5 rule. This allows the originating nation 5 QDUs, the international transit network 4 QDUs, and the terminating nation 5 QDUs. As the telecommunications market becomes more liberalised, a multi-operator environment makes the apportionment of a ‘nation’s’ QDU budget increasingly difficult. Clearly a GSM-to-GSM call, national or international, will either just meet or breach these guidelines. Furthermore, there is some subjective evidence to suggest that the older generation, who grew up with an analogue PSTN equipped with electro-mechanical switches, will accept the 14 QDU limit; however, the younger generation, who have only really known modern digitally switched fixed networks, will only accept an upper limit of 9 QDU.

Speech quality degradation can occur when a codec is tandemed with another, of either the same or a different type. Apart from the ITU-T G.721 [17] 32 kbit/s ADPCM standard, where synchronous coding adjustment allows ADPCM tandeming to occur without any further distortion to be incurred, most calls will incur more distortion when speech codecs are tandemed. This is where the QDU ceases to be an accurate measure as very low bit rate systems do not behave in a linear way because some of the speech data, necessary for the second codec to produce an accurate representation of the input speech, has already been removed by the first codec, thereby compounding the distortion effect. Although the ITU-T have now introduced the concept of impairment factors into the ITU-T G.113 Recommendation [16], there is some concern at present on the validity of these measures for planning the end-to-end distortion of connections with low bit rate systems. In addition, the standard only includes the impairment factors of ITU codecs. Tests would be required to assess the TCH-FS/HS and EFR codecs.

Clearly, any improvement in speech quality is to be applauded. The recent adoption of the US.1 algorithm as the GSM enhanced full rate (EFR) codec, has improved the speech quality of the GSM system, in error-free or low-error environments, to ‘wire-line’ quality levels. The challenge now is to further improve on this advance to make it as invisible to the customer as possible that the telephone they are using is a cellular radio.

In 1995 the European Telecommunications Standards Institute (ETSI) embarked on the development of an enhanced full-rate (EFR) GSM speech codec. It was felt that there had been sufficient advances in speech coding that a new codec could be developed that would significantly improve the perceived quality of the GSM system. After the performance criteria had been defined, a pre-selection test phase was undertaken where candidate codecs could be submitted, with subjective test results, and compared with the performance criteria. Two main candidates emerged, the US1 codec, developed by the Vocoder Task Group (TAG) of the North American PCS action group (NPAG) and the G.729EFR codec developed by BTL/Cellnet. TeleDanmark also submitted a G.729-based [18] solution with slightly different channel coding to that of the BTL/Cellnet candidate.

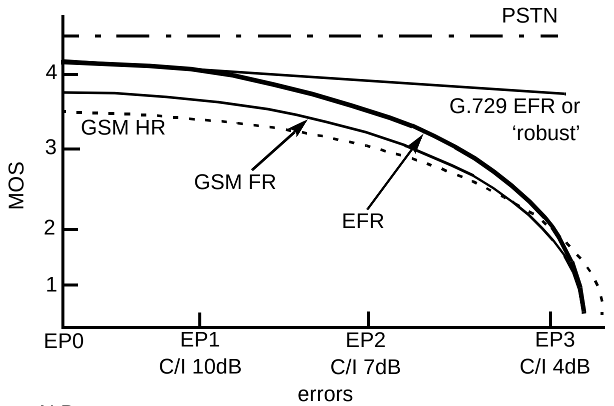
Most cellular network operators are currently embarked on major network expansion programmes aimed at providing high levels of handportable coverage and capacity. Even with the large numbers of cells currently foreseen there are still areas where the error performance of the system can be poor. For instance, radio planning tools rely on clutter databases to assess the building densities in a given geographic area, but their databases do not hold a picture of the actual buildings, rather an approximation of what is actually present; this is also true for terrain databases. The actual coverage will differ from the predicted coverage and, although drive testing can be used to check the quality and depth of coverage in a given area, it will not correct the entire network. The buildings that populate a given area all have different radio penetration losses because of the large variety of construction methods and even decor used, and this cannot be planned for by the network operator. Hence, the occurrence of medium (C/I 7 dB) and high (C/I 4 dB) error rates is more frequent than desired. One method of overcoming these problems is to examine the possibilities of ‘robust’ coding.

The current GSM full-rate speech codec can offer acceptable speech quality in low-to-medium error performance environments. It uses a 9.8 kbit/s channel codec to protect the speech frames which are broken down into class 1a, 1b and class 2 bits. The class 1a and 1b bits are the most important as they convey the coefficients for the speech decoders vocal tract filters.

The US1 codec has the advantage of an error free MOS of about 4.2 and used the existing GSM full-rate channel codec, facilitating ease of deployment at a low cost. The US1 codec also offers good ambient acoustic noise performance. The G.729EFR candidate offered a very robust solution, with ‘wire-line’ quality and good tandeming performance because the basic G.729 [18] codec at 8 kbit/s, with an error-free mean opinion (MOS) of 4.0, is able to use 14.8 kbit/s of the GSM 22.8 kbit/s full-rate channel for channel coding. Thus, while the US1 codec’s speech quality degrades at approximately the same rate as the GSM full-rate codec, the G.729EFR codec is able to maintain an MOS of 4.0 from error-free conditions through carrier-to-interference (C/I) ratios of 13 dB, 10 dB and 7 dB with only marginal degradation at C/I of 4 dB, in typical urban propagation environments. The G.729EFR codec has reduced ambient noise performance when compared to both the US1 codec and the GSM full-rate codec but, because of the robust channel coding, the ambient noise performance was constant across the range of radio conditions. ETSI adopted the US1 codec for GSM EFR as it offered improved quality with minimal deployment problems, as well as commonality with the US PCS1900 networks. ETSI have now mandated the investigation of the benefits and possible development of a ‘robust’ full-rate codec.

The G.729EFR codec technology has shown that a ‘robust’ codec is possible and that it can achieve ‘wire-line’ quality across a broad range of operating conditions. In theory it could offer additional capacity in the network as the limit of C/I used for radio planning could be relaxed. In practice, should a ‘robust’ codec be adopted, the slow associated control channel (SACCH) signalling channel becomes the limit of performance. The SACCH is about 2—3 dB more tolerant to C/I than the current full-rate speech codec. An additional 2—3 dB of C/I margin would provide operators with additional fringe coverage and greater depth of in-building coverage for speech services. A major gain in radio capacity and coverage could be achieved if the SACCH performance could be improved to the same levels as a ‘robust’ codec; this might provide an additional 5 dB or even 6 dB of C/I margin over the current full-rate system. Recently, papers presented within the GSM standards fora have indicated that additional SACCH robustness can be achieved by adjusting the SACCH ‘radio-link time-out’ parameter. The full radio benefits of a ‘robust’ codec would not be realised until the majority of the operator’s customer base was equipped with ‘robust’ mobiles, but the possibilities are worth exploring. The effect on GSM data services also needs careful consideration.

The G.729 [18] codec has shown the way to a ‘robust’ speech codec. Like the US1 codec, it is based on advanced code-excited linear prediction (ACELP) techniques and uses similar channel coding polynomials to the GSM half-rate speech codec. These common features will enable terminal manufacturers to exploit the possibilities for supporting multiple codecs on a given digital signal processor (DSP), where the amount of memory and the complexity of the codec are critical factors (see Fig 2 for relative performances).



N.B:
PSTN, G.721, G.728 [19] and G.729: only fixed network errors.
EP: error files, frequency hopping and typical urban propagation characteristics.

Fig 2 Relative performance of GSM speech codecs.

It may be possible for a ‘robust’ cellular codec to be used as the codec for satellite systems or even the universal mobile telecommunications system (UMTS), depending on the channel rate and hence the bit rate available for channel coding. It is more likely that a future enhanced half-rate (EHR) GSM codec would fulfil this role. Such a codec would have a bit rate of about 4 kbit/s and may align with the ITU-T 4 kbit/s speech codec development. Current industry consensus is that an EHR codec, even if it offered the same quality as the EFR codec, with no additional robustness, would not be available until beyond the year 2000.

BTL and Cellnet have recently proposed a method enabling a ‘robust’ codec to offer additional radio capacity to GSM cellular operators, over and above that gained by the significantly improved C/I performance. If the ‘robust’ codec is used in both a full-rate and a half-rate channel, it can offer, according to the channel conditions, an increase in capacity for the operator. Channel conditions are assessed at call set-up and, if the channel is good, a half-rate channel is allocated, if the channel conditions are poor, a full-rate channel is allocated. Should the half-rate channel degrade during the call, or indeed the full-rate channel improve, an intra-cell handover can be instigated to ensure the customer always receives acceptable quality. Since the codec is common to both channels, the basic speech quality, tandeming and noise performance are the same. Figure 3 illustrates the ‘common codec’ concept. Work is progressing to assess how robust the codec is in the half-rate channel and how fast the channel quality assessment and intra-cell handover can be performed. If the limit of acceptable performance is the EP1 (C/I 10 dB) error profile and all the customer base within the cell have a common codec, there is the potential to increase the traffic-carrying capacity from 8.2 to 14.9 Erlang for a two-carrier GSM BTS.

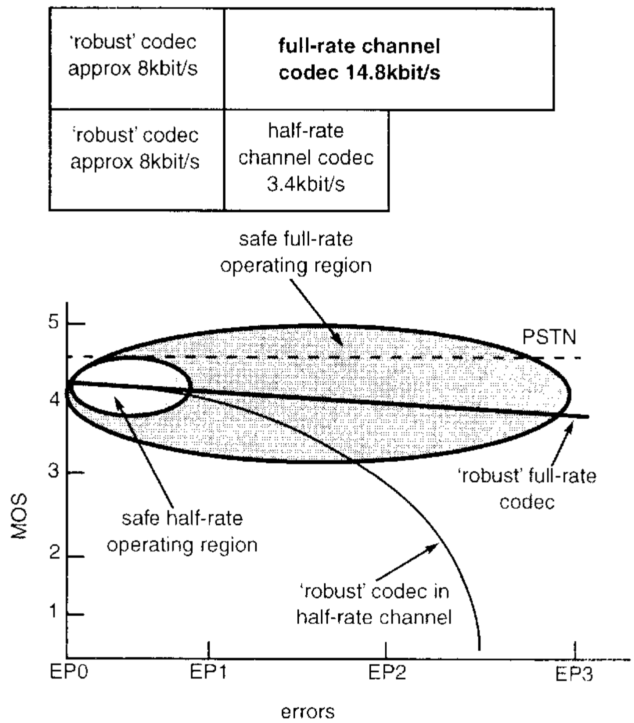


Fig 3 ‘Common codec’ concept for future GSM speech coding.

Discontinuous transmission (DTX) is another aspect of speech coding that affects the speech quality of the system. The DTX voice activity detector (VAD) is used to detect when the customer is speaking; this inevitably introduces some clipping which can reduce the speech quality. As acoustic noise also affects the DTX system, the full-rate VAD was optimised to work with in-car noise where the temporal characteristics are less dynamic than, say, street noise. As a result the VAD can be ‘falsely’ triggered, reducing the effectiveness of DTX at reducing C/I and extending mobile battery life.

The complexity of a given speech codec has a knock-on effect on the speech quality of the system. With a more complex algorithm, the speech codec can usually produce better speech quality for a given bit rate; however, to support that codec in mobile equipment, the signal processing technology has to execute the software in a reasonable period of time. Hence, complexity and delay are intrinsically linked. Excessive delay, as will be shown later, can seriously degrade the quality of a connection.

4. Echo control

The mobile switching centre (MSC) incorporates an echo canceller adhering to ITU-T G.165 [20] with a 60 ms echo path window, since the fixed network does not have echo control in the national network and the delay of the GSM system necessitates some form of echo control. It is important that the interaction of the MSC echo canceller with other network echo cancellers is understood. In an international connection between Europe and USA there will be an echo canceller in the home country's international switching centre (ISC) looking at the GSM network; this canceller will normally have a 64 ms echo path window but will be trying to cancel echoes over a 190 ms echo path. In addition, the nonlinear GSM speech codec renders the canceller ineffective and it should be switched out of the connection.

BTL have tested a number of commercially available echo cancellers in a variety of conditions, using the echo return loss enhancement (ERLE) method. The ITU-T G.165 [20] (blue book) uses white noise to test echo cancellers whereas the BTL test system uses segments of real speech and places the canceller in simulated network conditions, with delay, speech coding and other network equipment in the echo path. The test results were used to develop the 'correct echo control' feature for BT interconnect agreements with GSM mobile operators. Correct echo control defines the rules whereby the various echo cancellers in a connection are to be disabled or enabled.

The far-end ISC will have an echo canceller looking at the far-end customer; this is connected in tandem with the MSC echo canceller. The ERLE tests carried out by BTL have shown that echo cancellers connected in tandem can reduce the amount of echo control by only between 3 dB and 6 dB dependent on whether the centre clippers are active or not. For this reason the MSC echo canceller should be switched out. The mechanism for achieving 'correct echo control' uses the CCITT No 7 signalling system, acting upon the echo flag in the initial address message (IAM) and the final address message (FAM). In addition, when choosing an echo canceller, it is important to realise that most cancellers have been built to meet the blue book G.165 [20] which uses white noise to assess the canceller's performance; unfortunately, their performance can be reduced when operating with speech. Figure 4 shows a typical ERLE test results chart. Figure 5 illustrates the 'correct echo control' process.

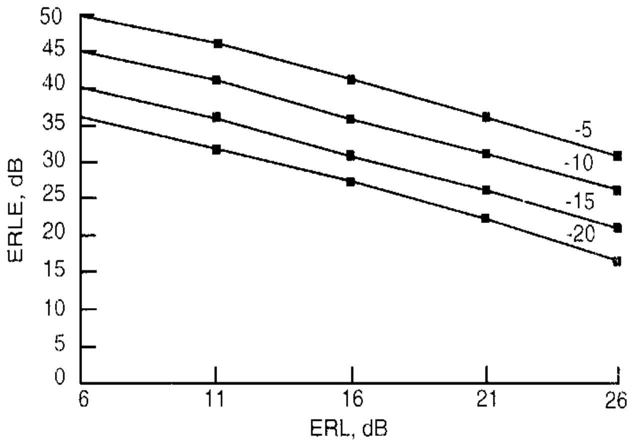


Fig 4 Typical echo return loss enhancement test results chart.

5. Fixed network performance

The fixed links used to connect the BTS to the BSC and on to the MSC via the TRAU can also affect the speech quality of the system. Most GSM operators employ submultiplexing to reduce their transmission costs; this allows the low bit rate speech to be multiplexed into a 64-kbit/s time-slot for onward transmission to the TRAU. Fixed link errors which occur here cause errors on the Abis speech frames which do not implement any error correction. Ultimately, this could cause a bad frame to be seen as good by the TRAU. The TCH-HS half-rate and TCH-EFR codecs do offer some Abis error protection but this is likely to involve a change to the BTS channel codec unit (CCU), involving some additional implementation cost.

It has been suggested that TRAU bypass should be implemented to eliminate the effects of tandemed codecs. It must be remembered that this will only be true for connections between mobiles on the same coding scheme, e.g. full-rate. It should also be remembered that the same speech frame will be subjected to two radio paths. Hence, a mobile in a low-error environment connected to another mobile in another low-error environment could create a high-error connection.

Codec tandeming is also a problem for non-real time communications. Voice messaging systems use low bit rate speech coding to store the message which adds another type of codec and more distortion to the connection. Call forwarding can also have an effect on the speech quality. Calls from a mobile to a fixed phone which has been diverted to a mobile will not be as good quality as a call which goes directly between the two mobiles as the echo control and delay will not be optimal.

Fixed link errors can have a dramatic effect on the signalling systems used between the BTS, BSC and MSC. Fixed networks are not as signalling intensive as mobile

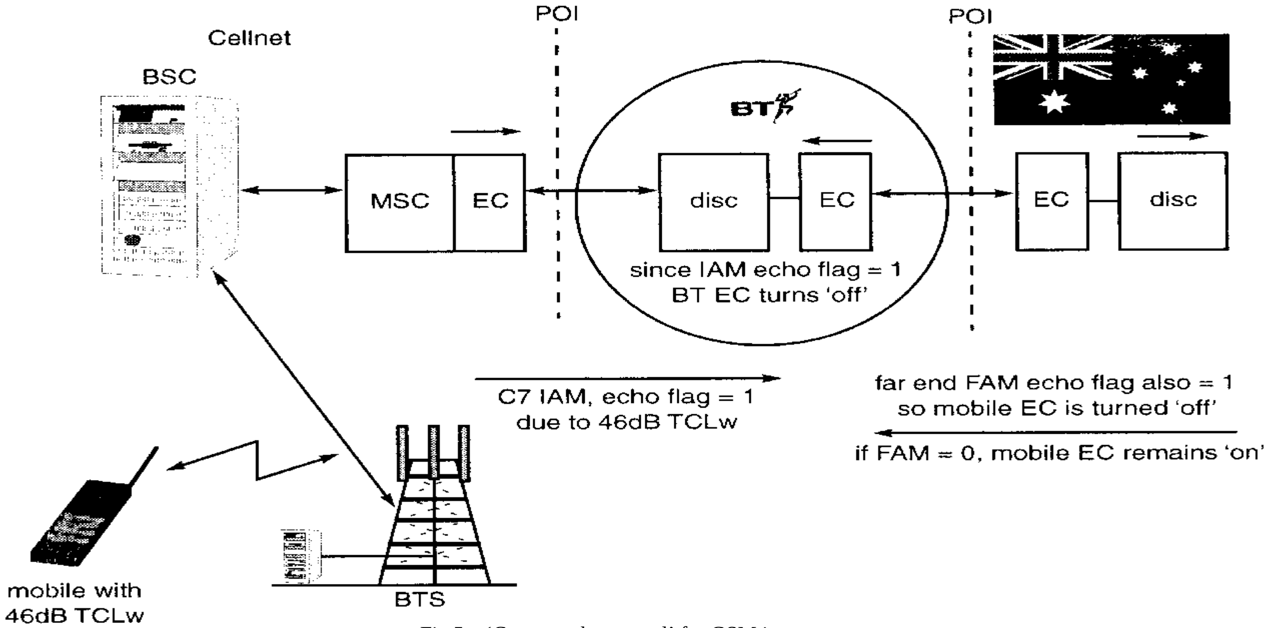


Fig 5 'Correct echo control' for GSM interconnect.

networks, where registration, authentication, location updating and short message services are performed as well as the more fundamental call set-up and clear-down messages. The signalling systems use a 'leaky bucket' error handling protocol which fills the bucket for every message received with errors, and empties the bucket for every n good messages received. If the fixed link bit error ratio is worse than 10^{-5} then the 'leaky bucket' fills very quickly, bringing the signalling link out of service and dropping all the calls in progress on that BTS or BSC.

When developing interconnect agreements it is important to remember the routeing of calls through the interconnected network. International calls, in particular, can be routed through digital circuit multiplication equipment (DCME) and digital speech interpolation (DSI) systems. These add distortion, delay and clipping which may be unacceptable when combined with the GSM system. The international network may make use of geostationary satellite routes which add a 260-ms one-way delay. GSM operators may wish to have their calls routed via cable connections as a preference. Indeed, BT offers a cable preference option for GSM interconnect which allows the GSM operator to request that GSM international calls are routed via cable (fibre) routes as a first choice and satellite as a second choice. Satellite avoidance cannot be guaranteed as some destinations can only be reached via satellite.

ITU-T G.114 [21], provides guidelines on delay limits for international circuits. It states that provided all other impairments are not present and echo control is provided, delays of up to 300 ms one way are acceptable. Above 300 ms the planner should exercise care as difficulties with conversation can occur. It is not recommended to exceed 400 ms. A GSM-to-GSM call routed via a satellite will incur 450 ms of delay and also includes nonlinear speech coding, radio error conditions and echo control. Network planners should be aware of the speech quality issues when designing the international routes for GSM calls.

6. End-to-end performance

When developing and implementing a new digital mobile system it is important that the effects of a given design on the end-to-end speech quality are understood such that a design that offers acceptable quality across the full range of call scenarios, is implemented. A balance must be found between the various engineering and commercial pressures that exist when designing the system.

Apportioning the various transmission parameters becomes increasingly difficult in a multi-operator environment. In many countries there is still only one fixed network operator with one or two cellular operators and a number of private branch networks (PBNs). With this scenario it is relatively easy to apportion delay, loudness ratings and the other transmission quality parameters; however, as telecommunications liberalisation progresses most countries will develop a multi-operator environment similar to that in the United Kingdom where there are over 150 licensed operators, offering local, trunk and radio networks and services.

Originally the United Kingdom relied on individual interconnect agreements between operators to agree the apportionment of transmission parameters across the network boundaries, guided by the network code of practice

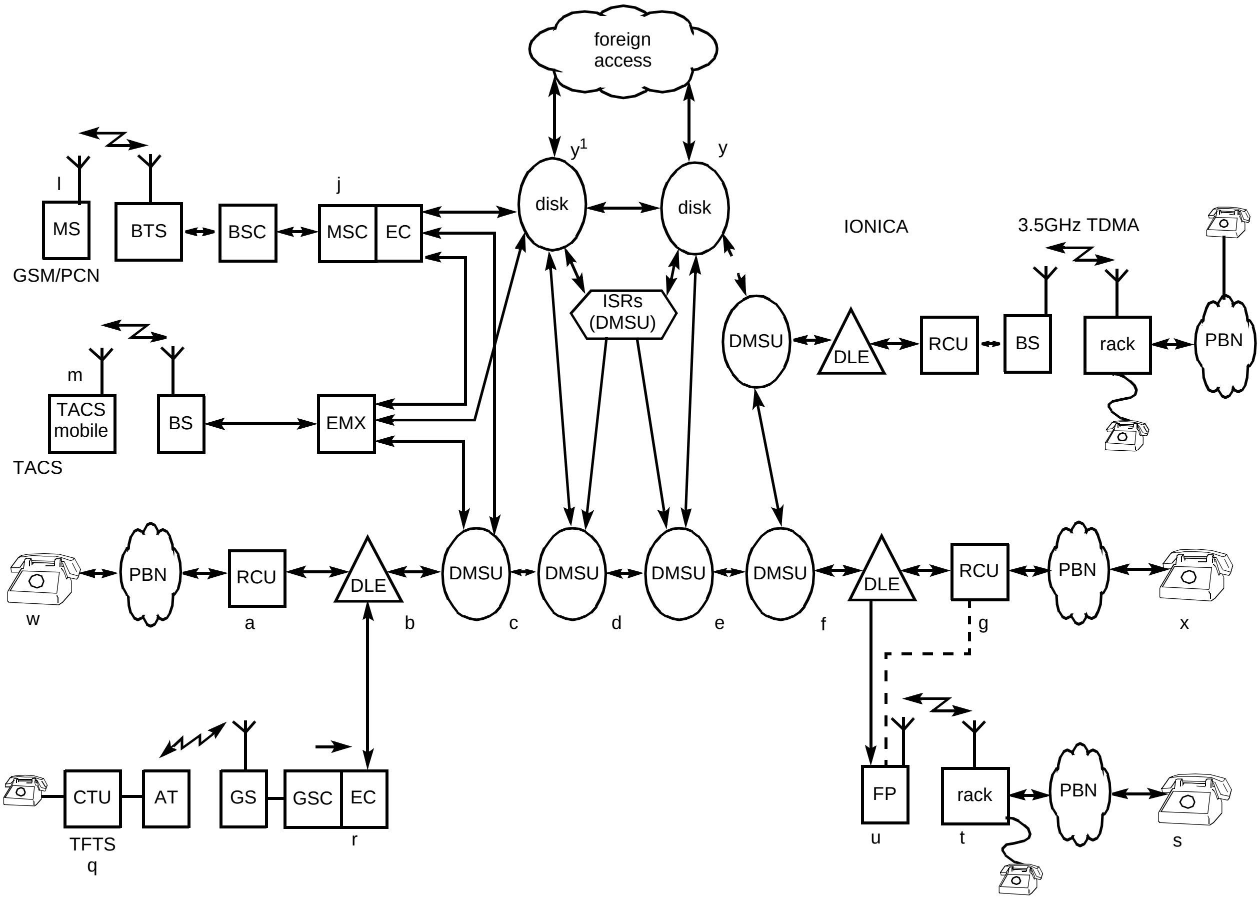


Fig 6 UK network model used for network performance design standards.

(NCOP). As the number of operators grew this technique became difficult to manage and the industry is now moving towards standard contracts with the transmission apportionment following the network performance design standards (NPDS) document [1] produced by the public network operators interest group (PNO-IG), a subgroup of the Department of Trade and Industry's (DTI) Network Interfaces Co-ordination Committee (NICC).

The NPDS document uses a network model to apportion transmission parameters. End-to-end limits are taken from ITU-T recommendations and individual system standards, such as the GSM recommendations. It is assumed that there are local loop operators, radio local loop (RLL) operators and trunk network operators, such as BT and Energis. Figure 6 shows the network model used to create the NPDS document [1].

One of the major principles of the document is that it is the responsibility of the operator that bills the customer to deal with any complaints. This means that if a fixed customer calls a mobile customer, hears echo and subsequently complains to the fixed operator, the fixed operator has the responsibility to address the problem with the mobile operator. Unfortunately, in this example, the echo is usually caused by the mobile customer's handset.

The only safeguard that the mobile network operator has that the handset will not cause echo is that it has passed a type approval test; this, as has alredy been shown, can prove inadequate. Current type approval test methods for digital cellular handsets do not fully represent their operational use and hence it is possible for handsets to pass the test but still cause echo while in use.

Mobile operators have little or no control over the handsets connected to their networks and the echo problems, if generated by a sufficient number of handsets, will affect a large number of calls. It is important, therefore, that the type approval tests continue to be carried out and that they reflect the operational use of the handset.

7. Conclusions

The GSM digital cellular system has become a global success. Further enhancements are possible to enable GSM to enter new markets and offer 'wireline' quality to customers, whether they require mobility or not.

As has been shown in this paper, ensuring the end-to-end speech quality of the GSM system is a matter of balance. The effect of any one parameter can have a consequent effect on other parts of the system and on interconnected networks, resulting in disputes and ultimately litigation.

Neither the terminal equipment nor the network can be expected to solve all the speech quality problems. A balance must be found between the needs of the network operators and the terminal equipment manufacturers to ensure that acceptable standards can be developed to the mutual benefit of the GSM community.

Some parameters are critical and should form the basis of a type approval system; however, not all parameters need to be type approved and the type approval system should not be designed to act as a barrier to equipment manufacturers. Currently the type approval system is the only guarantee that the network operators have that the terminals connected to their networks will meet the basic quality required.

Technology already exists that shows 'robust' speech coding is possible within a GSM environment. Development is continuing to progress this technology within the GSM standards bodies.

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Appendix

List of acronyms

ADPCM	adaptive differential pulse code modulation
AMPS	advanced mobile phone system
AT	aircraft termination
BSC	base station controller
BSS	base station sub-system
BTL	BT Laboratories
BTS	base transceiver site
C7	CCITT No 7 signalling
CCITT	Consultative Committee International for Telephony and Telegraph
CDMA	code division multiple access
C/I	carrier to interference
CT2	cordless telephone 2 (digital standard)
CTU	cabin termination unit
DCS1800	digital communications system 1800 (GSM at 1800 MHz)
DECT	digital European cordless telephone
DCME	digital circuit multiplication equipment
DISC	digital international switching centre
DLE	digital local exchange
DMSU	digital main switching centre
DTI	Department of Trade and Industry
DTX	discontinuous transmission
DSI	digital speech interpolation
EC	echo canceller
EFR	echanced full-rate
EHR	enhanced half-rate
EMX	electronic mobile exchange
EP	error profile (e.g. EP1, EP2 etc)
ERP	ear reference point
ETSI	European Telecommunications Standards Institute
FAM	final address message
FR	full-rate
GSC	ground switching centre
GSM	global system for mobile communications
HATS	head and torso simulator
HR	half-rate
IAM	initial address message
ITU-T	International Telecommunications Union — Telecoms
LRGP	loudness rating guard-ring position
MOS	mean opinion score
MRP	mouth reference point
MS	mobile station
MSC	mobile switching centre
NCOP	network code of practice
NICC	Network Interfaces Co-ordination Committee
NMT	Nordic Mobile Telephone
NPAG	North American PCS Action Group
NPDS	Network Performance Design Standard
PBN	private branch network
PCS1900	personal communications system 1900 MHz
PSTN	public switched telephone network
PNO-IG	Public Network Operators Interest Group
RCU	remote concentrator unit
RLR	receive loudness rating
SACCH	slow associated control channel
SLR	send loudness rating
TACS	total access communications system
TBR	technical basis for regulation
TCH-EFR	traffic channel — enhanced full-rate
TCH-FS	traffic channel — full-rate speech
TCH-HS	traffic channel — half-rate speech
TCL	terminal coupling loss
TCL_w	weighted terminal coupling loss
TFTS	terrestrial flight telephony systems
TRAU	transcoder rate adapter unit
VAD	voice activity detector

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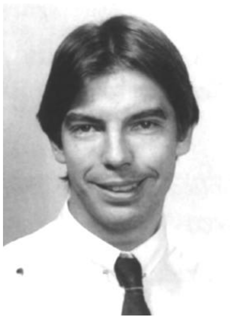
21 ITU-T G.114: 'Transmission impairments, delay', (1989).



Ian Goetz joined BT Laboratories in 1987 in the transmission performance unit, working in a team to develop a new type-approvals tester for ISDN telephones and a real time DSP system to model network speech paths for subjective and objective assessment. In 1990 he joined the cellular systems unit working on mobile transmission design issues. In this role he has represented BT on GSM, DECT and CT2 standards bodies and worked closely with Cellnet on transmission network design issues. In addition he filled the role of transmission and interconnect designer on the BT team that successfully bid for the second GSM licence in Spain. Recently he has represented BT and Cellnet on the GSM Speech Quality Strategy Group examining new speech codecs and quality issues for the GSM system. In 1996 he moved to join the Cellnet Technology Strategic Planning Group.